

N100 Financial Intelligence Platform

Analyst User Guide

Purpose

This guide explains how an analyst can use the N100 Financial Intelligence Platform to explore company financials, screen companies, compare peers, analyse sectors and trends, review capital structure information, and generate company reports.

Platform components

Component	Purpose
Streamlit Dashboard	Interactive analyst interface
FastAPI	Programmatic access to financial data
SQLite Database	Central financial data store
Analytics Engine	Ratios, clustering and financial analytics
Reports	Tearsheet and analytical outputs

1. Getting Started

The platform uses a SQLite database named nifty100.db. The interactive dashboard is implemented with Streamlit and the API layer is implemented using FastAPI.

Starting the API

```
.\venv\Scripts\python.exe -m uvicorn src.api.main:app --port 8000
```

The API is available at <http://127.0.0.1:8000>.

Starting the dashboard

```
streamlit run src\dashboard\app.py --server.port 8501
```

The dashboard is available at <http://localhost:8501>.

Recommended startup order

1. Activate the virtual environment.
2. Start FastAPI on port 8000.
3. Start Streamlit on port 8501.
4. Open the Streamlit dashboard in a browser.
5. Verify the API health endpoint if required.

2. Home Dashboard

The Home screen provides the starting point for analyst navigation. It is intended to provide a high-level view before moving into detailed company or analytical screens.

Typical workflow

Start with the overall market/company view, identify an interesting company or sector, and then move to the relevant detail screen.

Analyst use cases

- Discover companies for further research.
- Navigate to company profiles.
- Identify sectors requiring deeper analysis.
- Move from high-level information to detailed financial metrics.

The dashboard is designed so that an analyst can progressively move from overview information toward quantitative analysis.

3. Company Profile

The Company Profile screen is used for detailed analysis of an individual company.

Selecting a company

Select a company ticker such as TCS, INFY, HDFCBANK, RELIANCE or ITC. The profile screen retrieves company-specific financial information from the underlying database/API layer.

What to examine

- Company identity and sector.
- Historical financial ratios.
- Profitability measures.
- Capital structure.
- Cash-flow information.
- Historical trends.

The Company Profile is particularly useful when an analyst has already identified a company through the screener and wants to investigate it further.

4. Screener

The Screener allows analysts to filter companies according to financial criteria.

Available filtering concepts

The API supports filters including minimum ROE, maximum debt to equity, minimum free cash flow, sector, revenue CAGR, profit CAGR and maximum P/E.

Example

A request such as `/api/v1/screener?min_roe=15` returns companies whose return on equity is at least 15.

Analyst workflow

1. Set the desired financial filters.
2. Review the returned companies.
3. Identify candidates for further analysis.
4. Open individual Company Profiles.
5. Compare candidates with peer companies.

5. Peer Comparison

Peer analysis allows an analyst to compare companies operating within a relevant peer group.

Example API calls

```
GET /api/v1/peers/Private%20Banks
```

```
GET /api/v1/companies/TCS/peers/compare
```

The first request retrieves companies belonging to a peer group. The second provides a company-level comparison against the company's peer group.

Use peer analysis to determine whether a company's profitability, capital structure or other metrics are strong or weak relative to comparable businesses.

6. Trends and Sector Analysis

The Trends and Sectors screens provide broader analytical context beyond a single company.

Sector analysis

The platform contains sectors including Communication Services, Consumer Discretionary, Consumer Staples, Energy, Financials, Healthcare, Industrials, Information Technology, Materials and Real Estate.

For example, Information Technology contains companies such as HCL Technologies, Infosys, LTIMindtree, TCS and Tech Mahindra.

Analyst workflow

Compare sector-level profitability and then drill down to individual companies. Sector analysis is useful for identifying relative strengths and weaknesses across industries.

7. Capital Analysis

The Capital screen is intended to help analysts evaluate capital structure and financial strength.

Important metrics include debt to equity, interest coverage, free cash flow, cash generated from operations and other capital-related measures.

Interpretation

Lower leverage can indicate a stronger balance sheet, although appropriate leverage depends on the company's industry. Interest coverage provides an indication of the company's ability to service interest obligations.

Always interpret capital metrics alongside sector characteristics rather than using one ratio in isolation.

8. Reports and Tearsheet Generation

The Reports screen provides access to generated analytical outputs and company reports.

Tearsheets

Company tearsheets consolidate important financial information into a report suitable for analyst review.

Recommended workflow

1. Identify the company.
2. Review its Company Profile.
3. Review peer comparison.
4. Review financial and capital metrics.
5. Generate or open the corresponding tearsheet.

Generated reports should be checked for readability, complete content and absence of text overflow before final use.

9. API Usage

FastAPI exposes programmatic access to the platform's financial data and analytics.

Health check

```
curl http://127.0.0.1:8000/api/v1/health
```

Sector list

```
curl http://127.0.0.1:8000/api/v1/sectors
```

Sector companies

```
curl http://127.0.0.1:8000/api/v1/sectors/Information%20Technology/companies
```

Screener

```
curl "http://127.0.0.1:8000/api/v1/screener?min_roe=15"
```

10. API Endpoints and Data Access

The API includes endpoints for health checks, company data, sectors, screening, market-cap history, portfolio statistics, documents and peer analysis.

Endpoint	Purpose
/api/v1/health	API/database health
/api/v1/sectors	Sector-level information
/api/v1/sectors/{sector}/companies	Companies in sector
/api/v1/companies	Company listing
/api/v1/companies/{ticker}	Company profile
/api/v1/screener	Financial screening
/api/v1/portfolio/stats	Portfolio KPI statistics
/api/v1/peers/{group}	Peer group companies
/api/v1/companies/{ticker}/peers/compare	Peer comparison

11. Troubleshooting

Port 8000 already in use

Find the process using port 8000:

```
netstat -ano | findstr :8000
```

Terminate the process using its actual PID:

```
taskkill /PID <PID> /F
```

Python import errors

Set the project root as PYTHONPATH:

```
$env:PYTHONPATH = (Get-Location).Path
```

Testing

Run the complete test suite:

```
pytest tests\ -v
```

The project should be run from the repository root with the virtual environment activated.

12. Testing and Performance

The project includes automated tests covering ETL normalization, financial KPIs, data-quality rules, API endpoints and performance.

The completed test suite contains 92 tests, all of which passed during the Day 43 verification.

Performance results

The concurrent API load test completed 10 screener requests within approximately 1.28 seconds total, below the 10-second requirement.

The five sampled company profiles loaded substantially below the three-second requirement.

FastAPI and Streamlit were also verified running simultaneously on ports 8000 and 8501.

13. Analyst Workflow Summary

A recommended end-to-end research workflow is:

1. Open the dashboard.
2. Review the overall market/company information.
3. Use the Screener to identify candidate companies.
4. Open the Company Profile for detailed financial analysis.
5. Review Peer Comparison.
6. Examine sector and trend information.
7. Review capital structure and cash-flow metrics.
8. Generate or review the company tearsheet.
9. Use API endpoints when programmatic access is required.
10. Record conclusions and validate important figures before using them in investment or analytical decisions.

The platform should be treated as an analytical decision-support system. Financial metrics should be interpreted in context and not used as the sole basis for investment decisions.

End of Analyst Guide